UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Flowcharts and visuals

Procedures, decisions, concept maps and comparison tables.

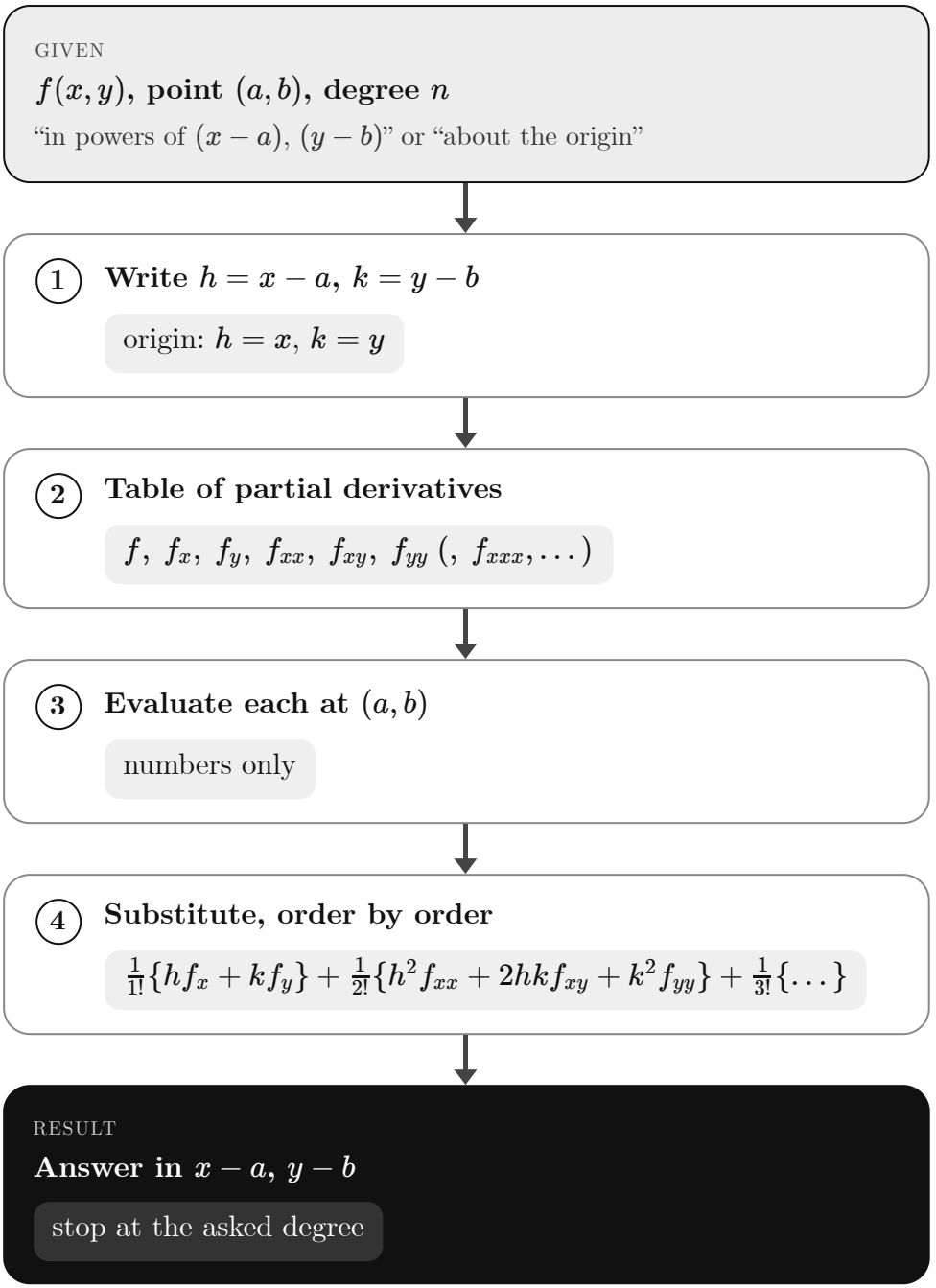
Class 007 · Taylor's and Maclaurin's expansion · compiled 14 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S007·9	Deck 007, slide 9 (rendered .pptx)
T007 00:18:07	AV-summary transcript 007, segment start
CW U1 P24	Unit 1 class-work problem 24
ESA Dec-23 1b	Past paper: Dec 2023, question 1.b

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3. Taylor's and Maclaurin's series side by side

3 Taylor's expansion of a function of two variables

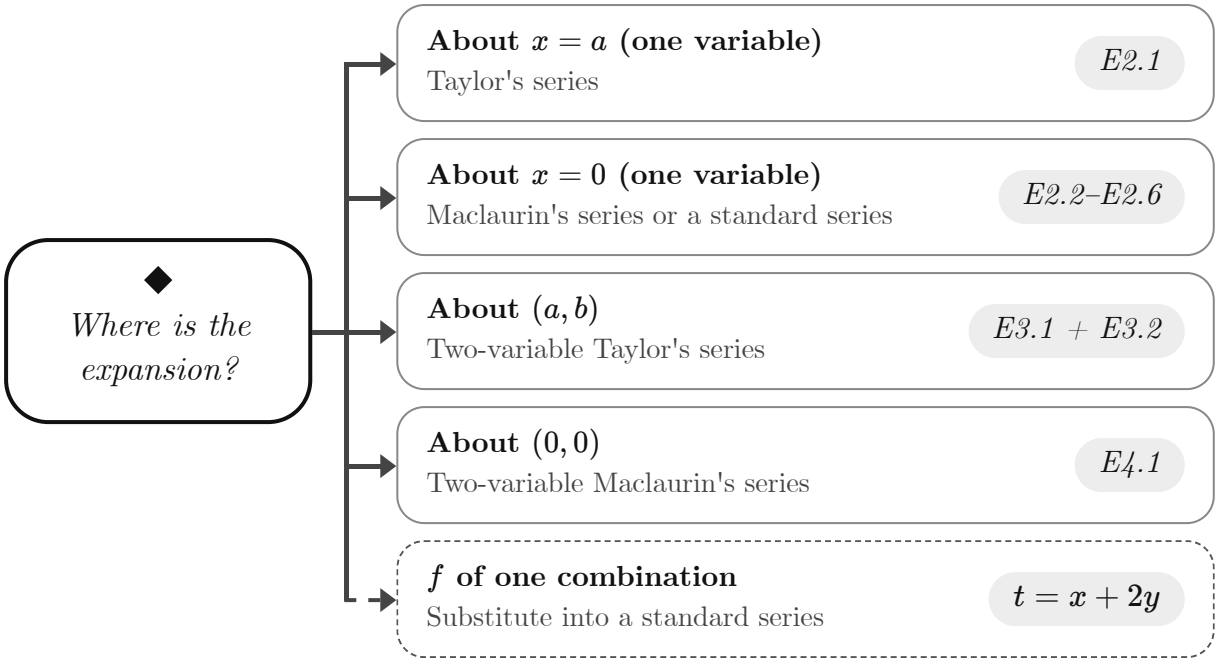


Remark. Check: if f depends only on $x + 2y$, $x - y$, ..., the one-variable series gives the same answer.

Figure F3.1. Expanding f(x, y) in Taylor's series ESA ★★★

S007·9 · ESA Jul-23 1b · ESA Dec-23 1b

4 Maclaurin's expansion of a function of two variables



Remark. The dashed route is a shortcut added for clarity; it is also the quickest way to check an answer.

Figure F4.1. Which series? ESA ★★

T007 00:11:17 · S007·9 · S007·10

Taylor's and Maclaurin's series side by side

	One variable	Two variables
Taylor (about a point)	$f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \dots$	$f(a,b) + \{hf_x + kf_y\} + \frac{1}{2!}\{h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}\} + \dots$
Maclaurin (about the origin)	$f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \dots$	$f(0,0) + \{xf_x + yf_y\} + \frac{1}{2!}\{x^2 f_{xx} + 2xyf_{xy} + y^2 f_{yy}\} + \dots$
First degree	Tangent line	Tangent plane
Coefficients at order n	$\frac{1}{n!}$	$\frac{1}{n!}\times$ binomial (1,2,1; 1,3,3,1)
Condition	f and derivatives finite at the point	f and all partials finite at the point

ADDED FOR CLARITY S007·4 · S007·9 · S007·10